

EXPERIMENT NO 1:

Investigating the significance of Characteristic Impedance (Z_0), and Propagation Constant (γ) of a Transmission Line using the following parameters a) Resistance b) Capacitance c) Inductance d) Conductance

Program:

```
EXP1-TRANSMISSION.sci ✕
1 clear; clc;
2 R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
3 w=2*pi*f;
4 Z=R+(%i*w*L);
5 Y=G+(%i*w*C);
6 Zo=sqrt(Z/Y);
7 C=round(real(Zo));
8 D=round(imag(Zo));
9 printf('-Zo=.%f+.j(%f) ohms\n',C,D);
10 P=sqrt(Z*Y);
11 a=real(P);
12 a1=round(a*10000)/10000;
13 printf('-Attenuation constant a=.%f neper/km\n',a1);
14 b=imag(P);
15 b1=round(b*10000)/10000;
16 printf('-Phase constant b=.%f radians/km',b1);
17
```

OUTPUT:

```
Scilab 2025.1.0 Console
-Zo = 14.000000 + j(-2.000000) ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
--> |
```

